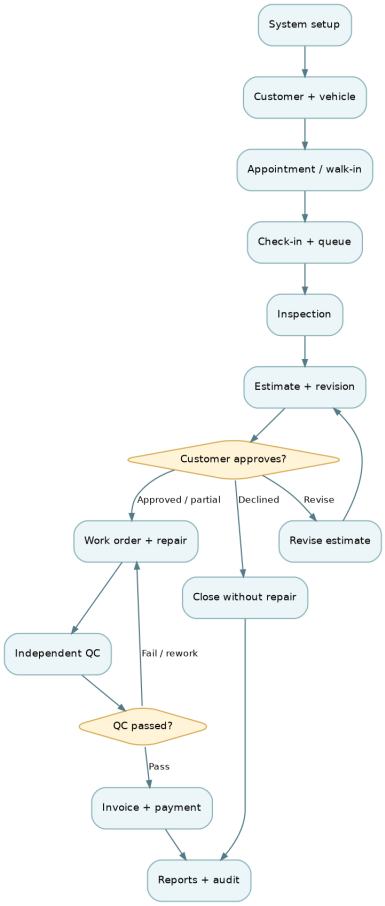
BACKYARDMECH

Complete System Master Plan

21 main system sections | 21 core workflows with rendered diagrams | 42 custom SQL Server tables and attributes | security architecture | business rules | implementation roadmap

Consolidated reference based on the latest agreed 42-table design. Agreed = previously planned; Recommended = implementation guidance; Pending = requires final design or enforcement.

System at a glance



PART I - THE 21 MAIN SYSTEM SECTIONS

1. Overall system architecture and flow

Customer Portal and Staff Management Portal communicate with an ASP.NET Core backend, ASP.NET Core Identity, authorization services, SQL Server, private inspection media storage, notification services, and audit logging. Main path: setup -> customer/vehicle -> appointment -> check-in -> inspection -> estimate/approval -> work order -> repair -> QC -> invoice -> payment -> reports. Declines, revisions, cancellations, and QC rework form exception paths.

2. Employee accounts, seven roles, and permissions

Roles: Shop Manager, Service Advisor, Mechanic, Parts Clerk, Tool Room Attendant, Quality Control Inspector, Accounting Clerk. Identity owns users/roles; EmployeeProfile stores employment information; Permission and RolePermission control module/action access. Backend authorization is mandatory.

3. Employee availability and schedule exceptions

EmployeeAvailability holds weekly working intervals; EmployeeAvailabilityException holds special availability and absences. Availability must subtract existing appointments and work-order allocations. Concurrent bookings require safe revalidation.

4. Garage master-data setup

Configure Bay, Supplier, PartCategory/Part, ServiceCategory/Service, ToolCategory/Tool, TaxRate, and ProductListing. Categories use self-referencing parent IDs, not separate subcategory tables.

5. Product listing

ProductListing is table 42 and publishes Parts or Services with title, description, image, display price, order, featured status, and visibility. Tool is internal garage equipment. Online cart and direct e-commerce checkout are not yet specified.

6. Customer registration

Customer stores personal and contact information. Staff should search for existing records before creating duplicates. Customer identity snapshots in historical transactions preserve traceability.

7. Vehicle registration and ownership transfer

Vehicle represents the physical car; VehicleOwnership preserves owner history with effective dates. VehicleOdometerHistory records mileage readings. Vehicle does not directly own a permanent CustomerId.

8. Appointment and walk-in registration

Both use Appointment with AppointmentType distinguishing entry path. AppointmentService allows multiple requested services. Preferred mechanic may differ from assigned mechanic. Check availability and conflicts.

9. Check-in and queue management

Record CheckedInUtc and staff snapshots on Appointment. Append QueueStatusHistory on status changes. Suggested states: Scheduled, Checked In, Waiting for Inspection, Under Inspection, Waiting for Approval, Ready for Repair, In Progress. Exact states remain pending.

10. Vehicle inspection

Inspection records mechanic, timing, and notes; InspectionMedia references photos/videos. Dedicated structured measurement/checklist rows are not present in the current 42-table schema.

11. Estimate creation and revisions

Estimate master -> EstimateRevision -> EstimateRevisionLineItem. Keep EstimateId stable, preserve historical revisions, record line-level approvals and revision-specific totals/tokens. Do not put EstimatedHours in estimate tables.

12. Customer approval and partial approval

Customer securely reviews exact revision and may approve all, approve some, decline, or request changes. Only approved scope may be executed. Superseded approval links should be invalidated.

13. Work order creation and assignments

WorkOrder links Appointment, Estimate, and ApprovedEstimateRevisionId; WorkOrderAssignment supports multiple mechanics. Bay/mechanic scheduling must avoid overlaps.

14. Actual repair and service time tracking

ServiceEntry stores actual service, mechanic, EstimatedHours, DurationMinutes, HourlyRate, ServiceCost. ServiceTimeSegment stores work/pause intervals. Actual worked time excludes pauses.

15. Parts inventory and supplier operations

Supplier -> Part; PartTransaction logs Received, Reserved, Installed, Returned, Wasted, Adjustment. Define reservation and on-hand accounting to avoid double deduction; negative inventory and concurrent overselling must be prevented.

16. Tool checkout, return, and maintenance

ToolCheckout tracks checkout/return timestamps, responsible mechanic, condition, and issues. ToolMaintenance records service/calibration. Unsafe or unavailable tools should not be issued.

17. Independent quality control

QualityCheck records QC result and inspector. QC inspector must differ from responsible repair mechanic. Failed QC triggers rework and another QC attempt; passing QC releases work for billing.

18. Invoice and tax calculation

Invoice and InvoiceLineItem represent billed actual services/parts. TaxRate has effective dates. InvoiceNumberCounter allocates sequential numbers transactionally. Planned statuses: Open, Paid, Void, Credited.

19. Payments, adjustments, credits, and voids

Payment stores amounts, methods, references, dates, and receiving employee; multiple payments can settle one invoice. InvoiceAdjustment preserves financial correction history. PaymentMethod is only on Payment.

20. Reports, dashboards, and audit history

Reports derive from operational tables: revenue, unpaid invoices, mechanic workload, service duration, stock, tool utilization, QC, queues, taxes. AuditLogEntry tracks important changes. No separate generic Reports or Transaction table.

21. System-wide business rules, history, and implementation requirements

Preserve snapshots and estimate revisions; prevent double-booking, unauthorized work, negative stock, overlapping service time, self-QC, duplicated invoice numbers, untraceable financial edits, and cross-customer data access. FK constraints, checks, indexes, security, and status transitions require implementation.

PART II - 21 DETAILED CORE WORKFLOWS

CORE FLOW 1. Employee account creation

Shop Manager creates employee -> Identity user -> EmployeeProfile -> role -> permissions -> weekly availability -> audit.



Security / integrity: Only authorized staff may create accounts or assign privileged roles.

CORE FLOW 2. Login and permission validation

Credentials -> lockout/rate limit -> optional/required MFA -> session -> role permissions -> record access -> execute and audit.



Security / integrity: Authorization must be enforced on the backend for each protected action.

CORE FLOW 3. Employee availability

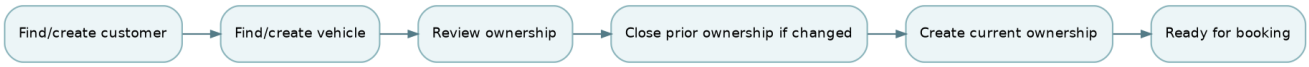
Requested slot -> weekly availability -> exceptions -> existing appointments/work orders -> bay availability -> recheck and confirm.



Security / integrity: Use transactional conflict checking for concurrent requests.

CORE FLOW 4. Customer and vehicle registration

Search customer -> create/select -> search vehicle -> create/select -> review ownership -> transfer if necessary -> vehicle ready.



Security / integrity: Verify customer authority over vehicle; preserve ownership history.

CORE FLOW 5. Product catalog publishing

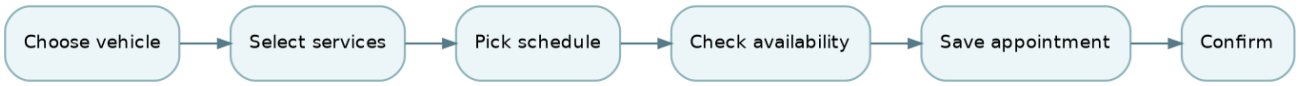
Create Part or Service -> optional ProductListing -> configure title, image, price -> publish -> customer browses -> requests service.



Security / integrity: Tools remain internal; never trust client display price for billing.

CORE FLOW 6. Appointment booking

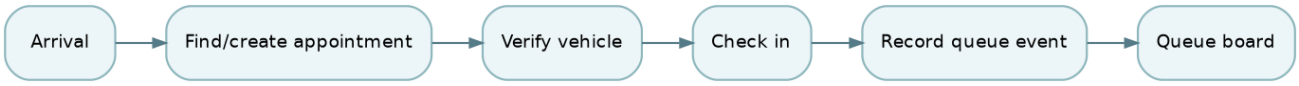
Choose services and vehicle -> date/time -> optional preferred mechanic -> availability -> create Appointment + AppointmentService -> confirmation.



Security / integrity: Validate customer/vehicle relationship and resource conflicts.

CORE FLOW 7. Walk-in and check-in

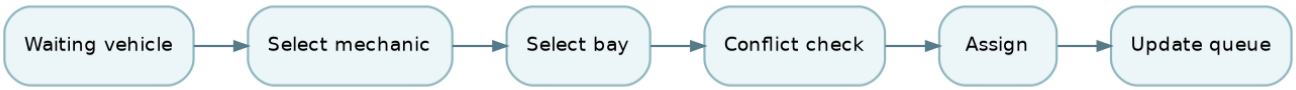
Scheduled or walk-in -> verify customer/vehicle -> check in -> CheckedInUtc -> QueueStatusHistory -> queue board.



Security / integrity: Record staff identity and timestamp for check-in.

CORE FLOW 8. Queue and bay allocation

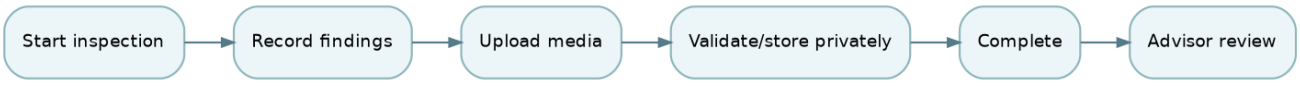
Waiting vehicle -> queue board -> mechanic/bay selection -> conflict validation -> assignment -> history -> inspection/work.



Security / integrity: Prevent race-condition double allocation.

CORE FLOW 9. Vehicle inspection

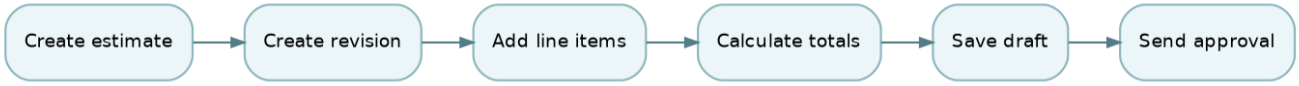
Begin Inspection -> findings/notes -> photos/videos -> secure upload -> complete -> advisor review -> estimate.



Security / integrity: Authorize file access and validate uploads.

CORE FLOW 10. Estimate creation

Inspection -> Estimate master -> revision -> parts/services -> server-calculated totals -> draft -> validate -> send.



Security / integrity: Server determines authoritative pricing/tax totals.

CORE FLOW 11. Estimate approval and revisions

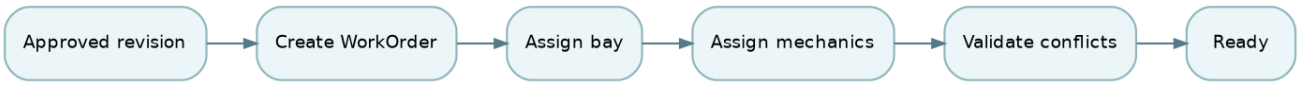
Customer opens secure token -> verify expiry/revision -> approve all/some/decline/request changes -> authorized scope or new revision.



Security / integrity: High-entropy expiring tokens; prevent replay and lock finalized revisions.

CORE FLOW 12. Work order creation

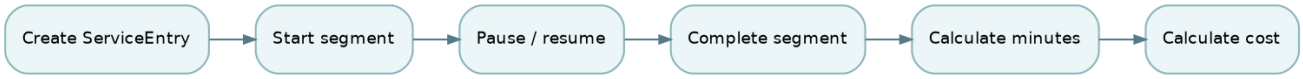
Approved scope -> WorkOrder referencing exact approved revision -> assign bay/mechanics -> conflict validation -> ready.



Security / integrity: Never include declined line items without new approval.

CORE FLOW 13. Service time tracking

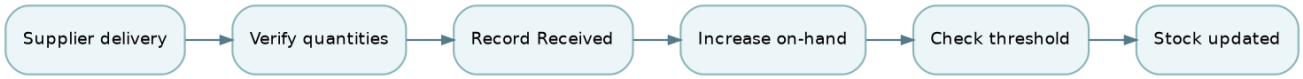
ServiceEntry -> start segment -> pause/resume -> finish -> sum work minutes -> compute cost -> complete.



Security / integrity: No overlapping/negative segments; audit post-completion changes.

CORE FLOW 14. Parts receiving and stock updates

Supplier delivery -> verify -> PartTransaction Received -> increase on-hand -> reorder check -> stock updated.



Security / integrity: Use atomic stock updates and restrict adjustments.

CORE FLOW 15. Parts reservation, installation and return

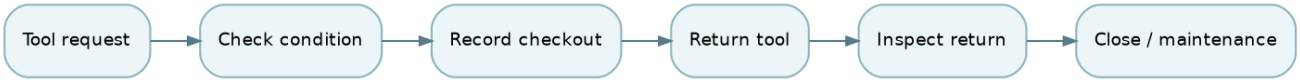
WorkOrder part request -> available stock -> reserve -> install/return/waste -> record transactions -> update inventory.



Security / integrity: Avoid double deduction and concurrent overselling.

CORE FLOW 16. Tool checkout and return

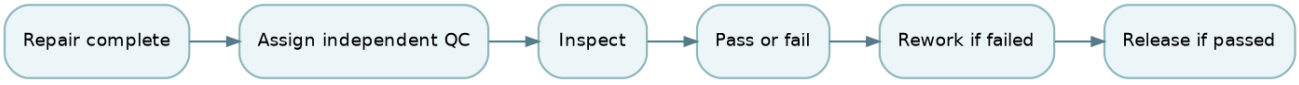
Request tool -> verify availability/calibration -> checkout -> use -> return -> inspect -> close or maintenance.



Security / integrity: Unavailable or unsafe tools cannot be issued without authorized override.

CORE FLOW 17. Quality control and rework

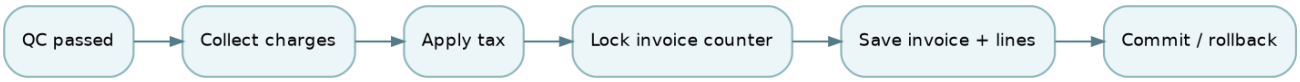
Completed repair -> independent inspector -> QC pass/fail -> rework on failure -> repeat -> billing on pass.



Security / integrity: Mechanic cannot approve their own repair.

CORE FLOW 18. Invoice creation

QC passed -> billable services/parts -> tax -> transaction -> lock counter -> invoice + items -> commit or rollback.



Security / integrity: Number allocation and invoice insertion must be one transaction.

CORE FLOW 19. Payment and adjustments

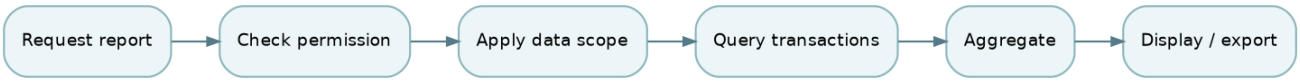
Open invoice -> validate payment -> save -> reconcile balance -> Paid/Open; correction -> authorized InvoiceAdjustment.



Security / integrity: Protect financial records from unauthorized edits and duplicates.

CORE FLOW 20. Reports and dashboards

Authorized report request -> permission/scope -> query transactions -> filters -> aggregate -> display/export -> audit sensitive exports.



Security / integrity: Prevent cross-customer access and unsafe exports.

CORE FLOW 21. Audit logging and incident response

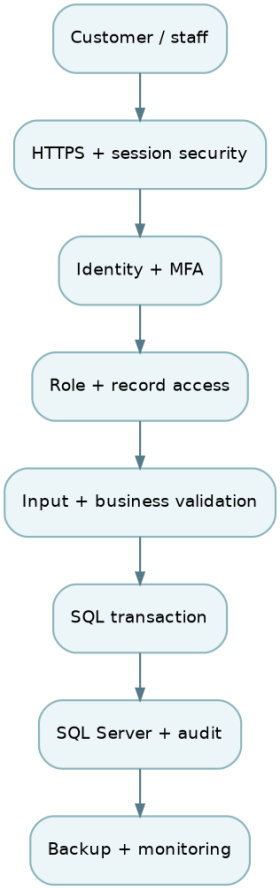
Event -> classify -> write audit/security log -> protect log -> monitor -> alert -> investigate -> recover and review.



Security / integrity: Restrict audit deletion/modification and avoid logging secrets.

PART III - SECURITY REQUIREMENTS

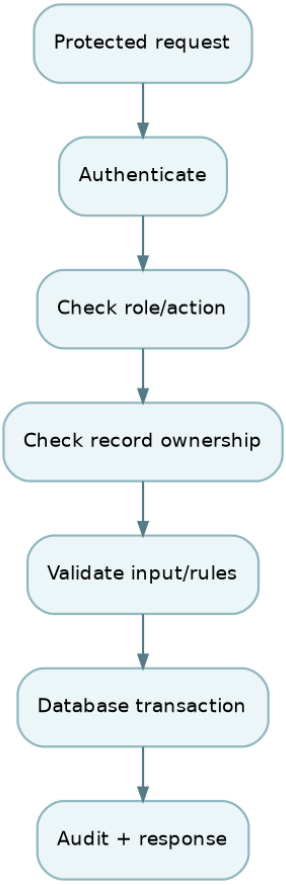
Security model: least privilege, defense in depth, zero trust of browser-submitted values, protected historical records, secure tokens, and auditable sensitive actions. The controls below are requirements/recommendations, not claims of completed implementation.



#	Measure	Expected implementation
1	Secure authentication	ASP.NET Core Identity; secure password hashing; no plaintext password storage.
2	MFA	Require MFA for privileged accounts; offer to other users.
3	Login abuse defense	Rate limits, temporary lockout, suspicious login monitoring.
4	Authorization	RolePermission and record-level authorization on each protected API.
5	Transport security	HTTPS everywhere, HSTS, secure cookies and appropriate SameSite.
6	CSRF protection	Antiforgery tokens for cookie-authenticated state-changing requests.
7	XSS prevention	Output encoding, safe HTML rendering, content security policy where practical.
8	SQL injection defense	EF Core/parameterized queries; no string-concatenated SQL.
9	Input validation	Validate types, ranges, lengths, enum values and business rules on server.
10	Secure uploads	Restrict size/content; validate file type; malware scanning where feasible.
11	Private media	Protect inspection files; authorized download or short-lived signed links.

#	Measure	Expected implementation
12	Approval token protection	Unpredictable high-entropy expiring tokens, hashed storage recommended, revocation and replay protection.
13	Encryption at rest	Database/storage encryption and encrypted backups where appropriate.
14	Secrets management	Use deployment secret stores; never commit keys/passwords to source control.
15	Audit logging	Capture actor, action, entity, time, and safe details for sensitive changes.
16	Log integrity	Restrict log alteration/deletion; retain under a documented policy.
17	Concurrency	Transaction-safe scheduling, stock reservation and invoice numbering.
18	Database least privilege	Use restricted app DB credentials; separate migrations/admin access.
19	Backup and recovery	Automated encrypted backups and periodic restore testing.
20	Monitoring and response	Detect suspicious activity; incident triage, containment and recovery.
21	Payment security	Do not store raw card data; use a compliant provider if gateway is introduced.
22	Privacy and retention	Data minimization, access restrictions, retention/disposal and Philippine DPA review.
23	Dependency security	Patch ASP.NET, SQL Server, libraries and infrastructure.
24	Security testing	OWASP ASVS baseline; IDOR, XSS, CSRF, injection, replay and concurrency tests.
25	Export security	Restrict financial exports; protect CSV/spreadsheet formula injection.
26	Financial integrity	Authorized credits/voids, non-reused issued invoice numbers, immutable history.
27	Service time integrity	No overlapping time segments; corrections logged.
28	QC separation	Enforce independent inspector for performed repairs.
29	Customer isolation	Customers cannot access others records by changing URL IDs.
30	Incident preparedness	Document contacts, recovery steps and evidence preservation.

Sensitive operation security sequence



Privacy: For Philippine deployment, review obligations under RA 10173 (Data Privacy Act of 2012) and applicable National Privacy Commission issuances. This blueprint is not a legal compliance certification.

PART IV - COMPLETE 42-TABLE ATTRIBUTE INVENTORY

1. EmployeeProfile

Attributes: EmployeeProfileId, UserId, EmployeeNumber, JobTitle, HourlyRate, HireDate, Phone, IsActive

2. EmployeeRateHistory

Attributes: EmployeeRateHistoryId, EmployeeProfileId, HourlyRate, EffectiveFrom, EffectiveTo, ChangedByUserId, ChangedByName, CreatedUtc

3. EmployeeAvailability

Attributes: EmployeeAvailabilityId, EmployeeProfileId, DayOfWeek, StartTime, EndTime, IsAvailable

4. EmployeeAvailabilityException

Attributes: EmployeeAvailabilityExceptionId, EmployeeProfileId, StartUtc, EndUtc, Reason, IsAvailable

5. Permission

Attributes: PermissionId, ModuleName, ActionName, PermissionCode, Description, IsActive

6. RolePermission

Attributes: RolePermissionId, RoleId, PermissionId

7. AuditLogEntry

Attributes: AuditLogEntryId, UserId, UserName, Action, EntityName, EntityId, TimestampUtc, Details

8. Customer

Attributes: CustomerId, FirstName, MiddleName, LastName, Email, Phone, StreetAddress, City, Province, PostalCode, Notes, CreatedUtc

9. Vehicle

Attributes: VehicleId, VIN, Make, Model, Year, LicensePlate, Color, Odometer, CreatedUtc

10. VehicleOwnership

Attributes: VehicleOwnershipId, VehicleId, CustomerId, OwnerFirstName, OwnerMiddleName, OwnerLastName, OwnershipStartUtc, OwnershipEndUtc

11. VehicleOdometerHistory

Attributes: VehicleOdometerHistoryId, VehicleId, AppointmentId, Odometer, RecordedUtc, RecordedByUserId, RecordedByName

12. Bay

Attributes: BayId, Name, Description, IsActive

13. Appointment

Attributes: AppointmentId, CustomerId, CustomerName, VehicleId, BayId, PreferredMechanicUserId, PreferredMechanicName, AssignedMechanicUserId, AssignedMechanicName, ScheduledStart, ScheduledEnd, AppointmentType, AppointmentStatus, CreatedByUserId, CreatedByName, CheckedInUtc, CheckedInByUserId, CheckedInByName, Notes, CreatedUtc

14. AppointmentService

Attributes: AppointmentServiceId, AppointmentId, ServiceId, ServiceName, Notes, CreatedUtc

15. QueueStatusHistory

Attributes: QueueStatusHistoryId, AppointmentId, QueueStatus, ChangedUtc, ChangedByUserId, ChangedByName, Notes

16. Inspection

Attributes: InspectionId, AppointmentId, MechanicUserId, MechanicName, StartedUtc, CompletedUtc, Notes

17. InspectionMedia

Attributes: InspectionMediaId, InspectionId, FileType, BlobUrl, UploadedUtc, UploadedByUserId, UploadedByName

18. Supplier

Attributes: SupplierId, SupplierName, ContactFirstName, ContactMiddleName, ContactLastName, Phone, Email, StreetAddress, City, Province, PostalCode, AccountNumber, IsActive

19. PartCategory

Attributes: PartCategoryId, ParentPartCategoryId, CategoryName, Description, IsActive

20. Part

Attributes: PartId, SupplierId, PartCategoryId, SKU, Name, Description, UnitCost, UnitPrice, QuantityOnHand, ReorderThreshold, IsActive

21. PartTransaction

Attributes: PartTransactionId, PartId, WorkOrderId, TransactionType, Quantity, UnitCostAtTransaction, PerformedByUserId, PerformedByName, PerformedUtc, Notes

22. ServiceCategory

Attributes: ServiceCategoryId, ParentServiceCategoryId, CategoryName, Description, IsActive

23. Service

Attributes: ServiceId, ServiceCategoryId, ServiceName, Description, DefaultPrice, DefaultEstimatedHours, IsActive

24. Estimate

Attributes: EstimateId, AppointmentId, CreatedByUserId, CreatedByName, CreatedUtc

25. EstimateRevision

Attributes: EstimateRevisionId, EstimateId, RevisionNumber, EstimateStatus, ApprovalToken, ApprovalTokenExpiresUtc, SentToCustomerUtc, RespondedUtc, Subtotal, TaxAmount, Total, RevisionReason, CreatedByUserId, CreatedByName, CreatedUtc

26. EstimateRevisionLineItem

Attributes: EstimateRevisionLineItemId, EstimateRevisionId, PartId, ServiceId, Description, LineType, Quantity, UnitPrice, LineTotal, ApprovalStatus

27. WorkOrder

Attributes: WorkOrderId, AppointmentId, EstimateId, ApprovedEstimateRevisionId, BayId, WorkOrderStatus, ScheduledStart, ScheduledEnd, ActualStart, ActualEnd, CreatedUtc

28. WorkOrderAssignment

Attributes: WorkOrderAssignmentId, WorkOrderId, MechanicUserId, MechanicName, AssignmentRole, AssignedUtc

29. ServiceEntry

Attributes: ServiceEntryId, WorkOrderId, ServiceId, MechanicUserId, MechanicName, ServiceDescription, EstimatedHours, DurationMinutes, HourlyRate, ServiceCost, Notes

30. ServiceTimeSegment

Attributes: ServiceTimeSegmentId, ServiceEntryId, SegmentStart, SegmentEnd, SegmentType

31. ToolCategory

Attributes: ToolCategoryId, ParentToolCategoryId, CategoryName, Description, IsActive

32. Tool

Attributes: ToolId, ToolCategoryId, Name, SerialNumber, Location, Condition, LastCalibratedUtc, NextCalibrationDueUtc, IsActive

33. ToolCheckout

Attributes: ToolCheckoutId, ToolId, WorkOrderId, MechanicUserId, MechanicName, CheckedOutUtc, ReturnedUtc, ConditionAtCheckout, ConditionAtReturn, ToolIssueStatus, Notes

34. ToolMaintenance

Attributes: ToolMaintenanceId, ToolId, MaintenanceType, PerformedUtc, NextDueUtc, PerformedByUserId, PerformedByName, Cost, Notes

35. QualityCheck

Attributes: QualityCheckId, WorkOrderId, InspectorUserId, InspectorName, Passed, PerformedUtc, Notes

36. TaxRate

Attributes: TaxRateId, Description, Rate, EffectiveFrom, EffectiveTo

37. Invoice

Attributes: InvoiceId, WorkOrderId, CustomerId, CustomerFirstName, CustomerMiddleName, CustomerLastName, InvoiceNumber, IssuedUtc, TaxRateId, Subtotal, TaxAmount, Total, InvoiceStatus, PaidUtc

38. InvoiceLineItem

Attributes: InvoiceLineItemId, InvoiceId, Description, SourceType, SourceId, Quantity, UnitPrice, LineTotal

39. InvoiceAdjustment

Attributes: InvoiceAdjustmentId, InvoiceId, AdjustmentType, Amount, Reason, PerformedByUserId, PerformedByName, PerformedUtc

40. Payment

Attributes: PaymentId, InvoiceId, Amount, PaymentMethod, ReferenceNumber, PaidUtc, ReceivedByUserId, ReceivedByName, Notes

41. InvoiceNumberCounter

Attributes: InvoiceNumberCounterId, Prefix, LastNumber

42. ProductListing

Attributes: ProductListingId, ListingType, PartId, ServiceId, ListingTitle, ListingDescription, ImageUrl, DisplayPrice, DisplayOrder, IsFeatured, IsPublished, CreatedUtc

Primary key for each table is its named *Id column. InvoiceNumberCounterId is not IDENTITY. The latest CREATE script has primary keys, defaults, and a unique InvoiceNumber but no foreign-key constraints. ASP.NET Identity tables are additional framework tables.

PART V - RELATIONSHIPS, VALIDATION, AND STATUS RULES

Relationship / rule	Implementation meaning
Customer -> VehicleOwnership -> Vehicle	Preserve owner history, no permanent Vehicle.CustomerId.
Appointment -> AppointmentService	One appointment can request multiple services.
Appointment -> Inspection -> InspectionMedia	Inspection findings and private photos/videos.
Appointment -> Estimate -> EstimateRevision -> LineItems	Stable estimate master and historical revisions.
WorkOrder -> ApprovedEstimateRevisionId	Exact approved version authorizes scope.
WorkOrder -> WorkOrderAssignment	Multiple assigned mechanics.
WorkOrder -> ServiceEntry -> ServiceTimeSegment	Performed service and time segments.
WorkOrder -> PartTransaction / ToolCheckout	Resources used in repair.
WorkOrder -> QualityCheck -> Invoice	Passed QC required before billing.
Invoice -> InvoiceLineItem / Payment / InvoiceAdjustment	Charges, receipts, and corrections.
Category parent IDs	Self-referencing hierarchies; prevent cycles.
ProductListing PartId / ServiceId	Exactly one source consistent with ListingType.
Part.QuantityOnHand	Never negative; stock movements transactional.
InvoiceNumber	Unique and allocated under transactional lock.

Suggested statuses (not finalized)

Entity	Proposed statuses
Appointment	Scheduled, Confirmed, CheckedIn, Waiting, InInspection, AwaitingApproval, ReadyForRepair, InProgress, Completed, Cancelled, NoShow
EstimateRevision	Draft, Sent, Viewed, Approved, PartiallyApproved, Declined, Expired, Superseded
WorkOrder	Draft, Scheduled, Ready, InProgress, Paused, AwaitingParts, AwaitingQC, ReworkRequired, Completed, Cancelled
Invoice (agreed)	Open, Paid, Void, Credited

Examples of constraints to add

Unique: Part.SKU, EmployeeProfile.EmployeeNumber, Permission.PermissionCode, (RolePermission.RoleId, PermissionId), (EstimateRevision.EstimateId, RevisionNumber). CHECK: positive Payment.Amount; nonnegative DurationMinutes and QuantityOnHand; ProductListing source type; valid date ranges. FKs: all relevant business relationships and category self-references. Implement authorization and conflict rules in services where simple constraints are insufficient.

PART VI - IMPLEMENTATION ROADMAP

Phase	Workstream	Deliverables
1	Foundation and security	Identity, permissions, employees, audit, FK/index/check design
2	Master data	Customers, vehicles, bays, suppliers, categories, catalogs
3	Customer experience	ProductListing, bookings, availability, walk-ins, check-in
4	Workshop intake	Queue, inspection, secure media
5	Estimates	Revisioning, secure approvals, partial decisions
6	Repair execution	Work orders, assignments, service timers
7	Resources	Stock reservations, installations, tool management
8	QC	Independent checks, failures, rework
9	Finance	Tax, invoices, counter, payments, adjustments
10	Release hardening	Reports, integration tests, security tests, backups, monitoring

PART VII - PENDING DESIGN / PRODUCTION GAPS

Area	Remaining work
Foreign keys and indexes	Not yet added to the latest CREATE script.
Inspection measurements	No dedicated structured checklist/measurement table yet.
Status transitions	Exact enums and allowed transitions remain to be finalized.
Scheduling conflicts	Need transactional overlap prevention and override policy.
Parts reservations	Need final reservation-balance and physical-stock accounting rules.
Estimate security	Token hashing/revocation, immutable revision approval, replay handling.
QC independence	Need enforced separation from responsible repair mechanics.
Invoice allocation	Need tested transaction-safe numbering and preservation of voided numbers.
Payment policy	Refund, reversal, overpayment and reconciliation rules.
Product checkout	Catalog browsing planned; full shopping cart not designed.
SRS alignment	Older names/relationships should be updated to latest 42-table design.

Final master flow

Secure setup -> customer/vehicle -> catalog/appointment/walk-in -> check-in/queue -> inspection -> estimate/revision -> customer approval -> approved work order -> mechanic/bay assignment -> actual repair and service timing + parts + tools -> independent QC (rework if failed) -> invoice -> payments/adjustments -> reporting, history, and audit.

This document is a consolidated project blueprint, not evidence that the controls or SQL constraints have already been implemented.